

Artificial Intelligence Mock Exam I

Time: 120 minutes

Total: 100 points

Language: Questions are in English. Answers may be written in English or Chinese.

Notes: Calculator allowed.

Part I. Multiple Choice Questions

Choose the best answer for each question. Each question is worth 2 points.

1. Which statement best describes the Markov property?

- A. The next state depends on all previous states and all previous actions.
- B. The next state depends only on the current state and action.
- C. The reward depends only on the initial state.
- D. The policy must be deterministic.

2. In fuzzy logic, if

$$\mu_A(x) = 0.3, \quad \mu_B(x) = 0.8$$

and the definition of implication is the same as that in propositional logic.

Then $\mu_{A \rightarrow B}(x)$ equals

- A. 0.3
- B. 0.7
- C. 0.8
- D. 1.1

3. Which of the following is a Horn clause?

- A. $P \vee Q$
- B. $\neg P \vee \neg Q$
- C. $\neg P \vee \neg Q \vee R$
- D. $P \vee Q \vee R$

4. In CLIPS, which construct is most directly used to declare a structured fact with named slots?

- A. `defrule`
- B. `deftemplate`
- C. `defmodule`
- D. `def facts`

5. In a nonstationary multi-armed bandit problem, which update is usually more suitable?

- A. Exact average with step size $1/n$
- B. Constant step size update

- C. A fixed policy without any update
- D. Breadth-first search

6. Which of the following models can learn bidirectional context representations?

- A. BERT.
- B. GPT.
- C. Qwen.
- D. All of the above.

7. Why is Layer Normalization often more suitable than Batch Normalization in sequence models such as Transformers?

- A. It needs larger convolution kernels.
- B. It does not depend on batch statistics.
- C. It requires labeled data.
- D. It removes the need for backpropagation.

8. In a knowledge graph, How many tuples do we usually use to represent knowledge?

- A. 1
- B. 2
- C. 3
- D. 4

Part II. Problems

9. Logic, CNF, and Resolution (16 points)

Consider the following statements.

1. Every AI student who studies logic writes some program.
2. Every program is either a Prolog program or a CLIPS program.
3. Anyone who writes a Prolog program can solve symbolic problems.
4. Anyone who writes a CLIPS program can solve symbolic problems.
5. Anyone who can solve symbolic problems passes the logic assignment.
6. Ming is an AI student and studies logic.

Do the following:

- (1) Translate the six statements into first-order logic. (6 points)
- (2) Negate the conclusion "Ming passes the logic assignment" , and convert the whole problem into clause form. (5 points)
- (3) Use resolution to prove that Ming passes the logic assignment. (5 points)

10. Rule-Based Expert System (10 points)

Consider the following CLIPS templates:

```
(deftemplate course
  (slot name)
  (slot compulsory))
```

```
(deftemplate score
  (slot student)
  (slot course)
  (slot value))
```

```
(deftemplate warning
  (slot student)
  (slot course)
  (slot reason))
```

```
(deftemplate passed
  (slot student)
  (slot course))
```

Write CLIPS rules for the following tasks:

- (1) If a student's score in a course is at least 60, assert a passed fact. (3 points)
- (2) If a student's score in a compulsory course is below 60, assert a warning fact whose reason is fail-compulsory. (4 points)
- (3) Explain briefly why binding the score fact address is useful if we later want to modify or retract that fact. (3 points)

11. Hidden Markov Model and Viterbi Algorithm (14 points)

Consider the hidden Markov model

$$\lambda = (A, B, \pi)$$

with

$$A = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \\ 0.3 & 0.7 \end{bmatrix}$$

$$\pi = (0.5, 0.3, 0.2)^T$$

The hidden states are $\{s_1, s_2, s_3\}$ and the observation symbols are $\{red, white\}$.

Given the observation sequence

$$O = (white, red, white)$$

compute the best hidden state path by the Viterbi algorithm.

12. Forward and Backward Computation in a Neural Network (14 points)

Consider a neural network with one hidden layer and no bias. The input is

$$x = [2, 1, 1]^T$$

The hidden layer uses ReLU, the output layer also uses ReLU, and the loss is

$$L(y, t) = \frac{1}{2}(y - t)^2$$

The weights are

$$W = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$V = [3 \quad -1]$$

and the target is

$$t = 2$$

Do the following:

- (1) Compute the hidden representation h and the output y . (4 points)
- (2) Compute the loss L . (2 points)
- (3) Compute

$$\frac{\partial L}{\partial V}$$

and

$$\frac{\partial L}{\partial W}$$

by backpropagation. (6 points)

(4) Is there a dead hidden neuron in this example? Explain briefly. (2 points)

13. Reinforcement Learning (14 points)

Consider an episodic MDP with discount factor

$$\gamma = 0.9$$

There are two nonterminal states s_1 and s_2 , and one terminal state T .

At s_1 , policy π chooses action a with probability 0.7 and action b with probability 0.3.

Transitions:

- From s_1 with action a : go to s_2 with reward 2.
- From s_1 with action b : go to T with reward 0.
- At s_2 , there is only one action c , which goes to T with reward 3.

Do the following:

- (1) Compute $v_\pi(s_2)$ and $v_\pi(s_1)$ using the Bellman equation. (6 points)
- (2) Suppose the current action-value estimates are

$$Q(s_1, a) = 1.2, \quad Q(s_2, c) = 2.0, \quad Q(s_2, d) = 1.5$$

and a transition

$$(s_1, a, r = 2, s_2)$$

is observed. Assume that under the current behavior policy, the next action actually chosen at s_2 is d . Let

$$\alpha = 0.1, \quad \gamma = 0.9$$

Compute one Sarsa update and one Q-learning update for $Q(s_1, a)$. (4 points)

- (3) Explain in one short paragraph why the two updated values in (2) are different. (4 points)

14. Transformer Positional Encoding (8 points)

In sinusoidal positional encoding, for a fixed dimension pair indexed by frequency ω , define

$$PE_{pos} = \begin{bmatrix} \sin(\omega pos) \\ \cos(\omega pos) \end{bmatrix}$$

Show that for any fixed offset k , there exists a 2×2 matrix M_k such that

$$PE_{pos+k} = M_k PE_{pos}$$

Write down M_k explicitly and explain why this property is useful.

15. Genetic Algorithm Design (8 points)

Design a genetic algorithm to solve the following problem.

There are m treasures with values

$$v_1, v_2, \dots, v_m$$

and they need to be split into 3 groups so that the total values of the three groups are as balanced as possible.

Describe clearly:

- (1) how to encode an individual,
- (2) how to define the fitness function,
- (3) how to perform initialization, selection, crossover, and mutation,
- (4) what stopping criterion you would use.